Chapter 16 exercises

DM16-E01

Decide each statement with conditions and witnesses. Is $GL_1(\mathbb{R})$ abelian? Is $GL_2(\mathbb{R})$ abelian? If not, exhibit two invertible noncommuting matrices. For $n \geq 2$, is $(\mathbb{Z}_n, \cdot)$ a group? For prime p, which residue subset forms a multiplicative group?

DM16-E02

Prove uniqueness of group identity and inverse, then prove left cancellation. Find the order of 3 in $(\mathbb{Z}_{12}, +)$.

DM16-E03

For $\varphi : \mathbb{Z}_{18} \to \mathbb{Z}_6$, $[k]_{18} \mapsto [k]_6$, prove representative independence and the homomorphism property. Find kernel, image, and the quotient's structure.

DM16-E04

In $(\mathbb{Z}_{12}, +)$, take $H = \{0, 4, 8\}$. List distinct left cosets, prove they partition the group, and compute the index using Lagrange.

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DM16-E05

Let $G = S_3$, $H = \langle (12) \rangle$, $g = (13)$. List gH, Hg, determine normality, and use two representatives of the same coset to show how proposed quotient multiplication can depend on representatives. Compose permutations right to left.

DM16-E06

Prove both directions: for $H \leq G$, multiplication $(aH)(bH) = (ab)H$ is well defined for all representatives if and only if $H \trianglelefteq G$.

DM16-E07

Take $p = 5, q = 11, e = 17$. Find $n, \varphi(n), d$, and encrypt and decrypt $M = 7$, displaying intermediate modular squares.

DM16-E08

Use the same keys with $M = 15$, which is not coprime to n. Compute encryption and decryption completely with CRT. Do not apply Euler's theorem without its coprimality condition.

DM16-E09

For $C = \{000, 111\}$, state detection and correction capabilities on a binary symmetric channel. If 111 is sent and 100 received, analyze nearest decoding and why correcting one error does not simultaneously guarantee detection of every two-error event under that decoder.

DM16-E10

Find all three pairwise distances for $x = 01010, y = 11001, z = 00011$ and check the triangle inequality. Compute $V(5, 2)$. How many errors does distance 5 guarantee can be corrected?

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DM16-E11

Work over $\mathbb{F}_2$ with

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Verify $GH^T = 0$, encode message 1010, and prove minimum distance 3 using the columns of H.

DM16-E12

For $C = \{000, 011, 101, 110\}$, $H = [1\ 1\ 1]$, list its two cosets, syndromes, and every possible coset leader. Does it uniquely correct every single-bit error?

DM16-E13

Given codeword $c = 1010101$ of Problem 11 and received word $r = 1110101$, use the syndrome to locate and correct the error. Verify from sphere packing that the $[7, 4, 3]$ code is perfect for one-error correction.

DM16-E14

Color four square vertices with three named colors and act by all eight symmetries D_4. List fixed counts by motion type and find the orbit count. Explain the denominator through counting fixed pairs in two ways.

DM16-E15

Give cycle indices for the natural square-vertex actions of C_4, D_4, and evaluate their three-color orbit counts.

DM16-E16

For a three-position rotational necklace C_3, use colors r, w, b to derive the full pattern inventory. Explain $2rwb$, and check its total coefficient sum against Burnside.

DM16-E17

Prove or refute each statement: (a) $GL_1(\mathbb{R})$ is nonabelian. (b) Multiplication on all of $\mathbb{Z}_p$, including zero, is a group for prime p. (c) Every subgroup automatically defines quotient group multiplication. (d) Zero syndrome proves transmission had no error. (e) Burnside's denominator is $|X|$.

DM16-E18

List every additive homomorphism $\mathbb{Z}_{18} \to \mathbb{Z}_{12}$. Give $\varphi(1)$, kernel, image and surjectivity for each. The natural reduction map is not well defined for these moduli.

DM16-E19

For $m, n \geq 1$, prove that $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic if and only if $\gcd(m, n) = 1$. Compare $\mathbb{Z}_6 \times \mathbb{Z}_3$ with $\mathbb{Z}_{18}$. In $\mathbb{Z}_{18}$, list every generator and every solution of $6x = 0$.

DM16-E20

Extend every word of this chapter's $[7, 4, 3]$ code by the parity bit $c_8 = \sum_{i=1}^{7} c_i \pmod 2$. Prove the new dimension and minimum distance. Let s be the original syndrome on the first seven positions and p the total parity of all eight. Assuming at most two errors, give the decoder decisions for all four combinations of zero or nonzero s and zero or one p. Which decisions lose their guarantee if this assumption is removed?

DM16-E21

Color six labeled circular positions red or blue, with exactly three positions of each color. Count orbits under C_6 rotations and under D_6 rotations and reflections by extracting coefficients from cycle types. Explain why directly dividing $\binom{6}{3}$ by six fails.

DM16-E22

Let $C_4 = \langle r \rangle$ act on the two diagonal vertex sets $P = \{\{1, 3\}, \{2, 4\}\}$. Color each whole set red or blue. List all four fixed counts, count orbits and find the stabilizer of a configuration with one red set and one blue set. Does Burnside still apply when an action is not faithful, so distinct group elements can induce the same permutation? Explain.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 16.

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